

should probably split the money into various piles and certainly not restrict yourself to the all-or-nothing bank deposit or stock market choice.

But let's examine the all-or-nothing question in greater detail, focusing on the two extreme alternatives. If you "take the plunge" and invest the \$10,000 in the stock market, what are the odds that you will regret this decision? What is the probability that the stock market is the wrong place to invest your money?

In a sense, you may have noticed, those questions are really pointless. They are pointless because no meaningful answer can be determined unless we specify an appropriate investment holding period. If you take the plunge, will you regret the decision tomorrow? Will you regret the decision in one year? Will you regret the decision in 10 years? When do you need the money?

Figure 3.3 illustrates what I call the time-adjusted "probability of financial regret." Over a 1-year time horizon, it shows that there is a 35% chance that a diversified portfolio of U.S. equities will under-perform the 5% rate of return from the safe bank deposit. What about a 10-year horizon? In that case, the chart shows that a diversified portfolio of U.S. equities has just an 11% probability of shortfall. In other words, there's a better than four-in-five chance that, over the next decade, the U.S. stock market will experience a rate of return that is greater than the return currently available from a secure 5% bank deposit.

Finally, what if your time horizon is 35 years away? In this case, the chart indicates a probability of financial regret of roughly 1%. As the time horizon increases, the probability of shortfall decreases rapidly; scientists like to say that it decays exponentially. It never actually hits zero—there are no guarantees in life—but it gets very close to zero. Time and financial risk are intricately intertwined.

Just as we must know the appropriate unit of time to comprehend a given speed, financial risk has an embedded dimension of time. Talking about whether something is risky or safe without addressing the relevant time horizon and the financial alternatives is meaningless. Indeed, over a 1-year horizon, the stock market is clearly quite risky—relative to the secure alternative of putting the money in a CD. That's because the probability of shortfall, or regret from investing in stocks, is 35%.